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IGNITING IDEAS



RIA & SIMNA

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Project Objective

- Familiarization of RISCV
- Design and verification of RISCV based SoC

What is RISC-V?

- A new free and open ISA developed at UC Berkeley starting in 2010
- Designed for
  - research
  - education
  - commercial use

Features of RISC-V



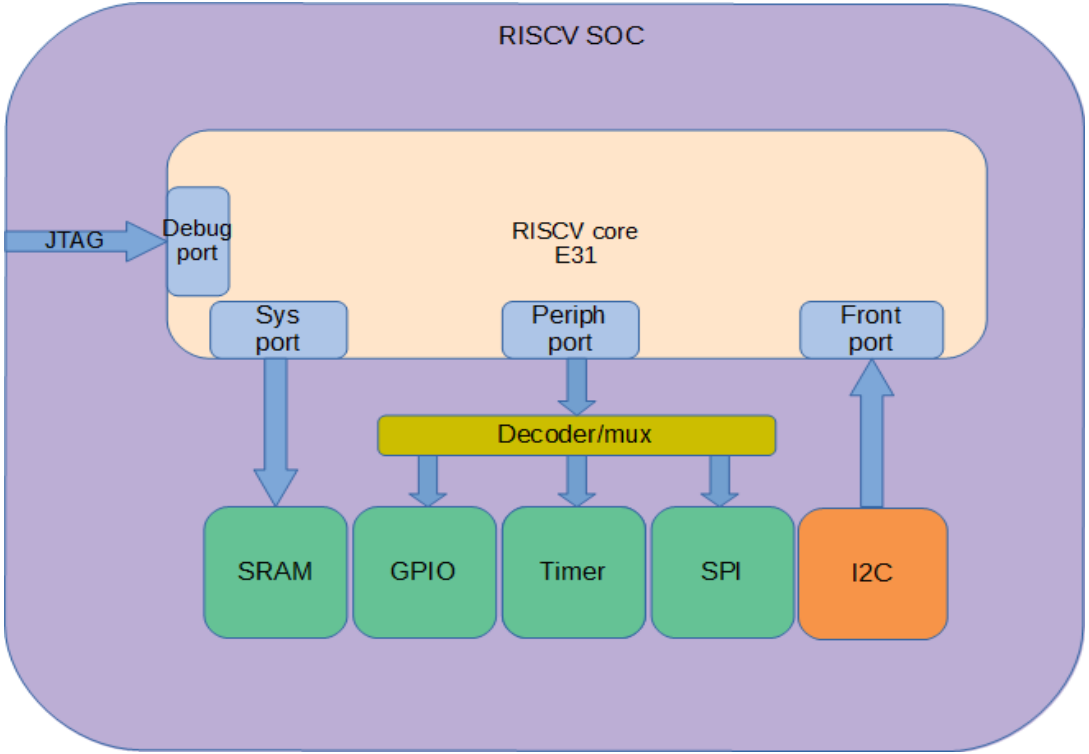
- Free and open
- Fixed Instruction Set Architecture (ISA)
- 64- bit and 32-bit variant
- Compressed Instructions support
- Separate privileged ISA
- Floating point standard support



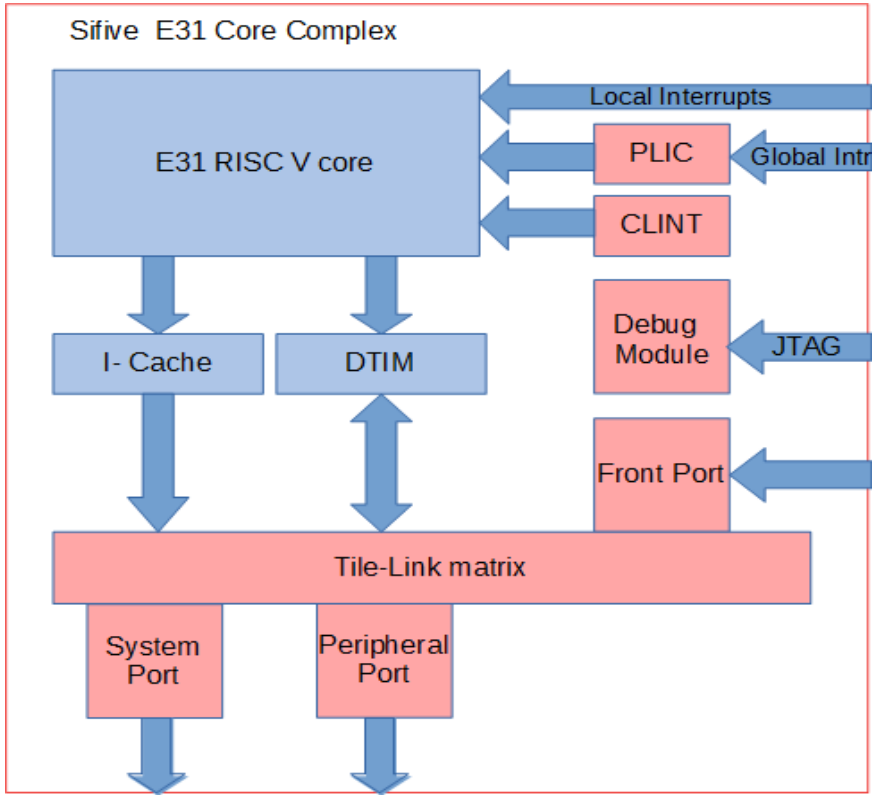
DESIGN



BLOCK DIAGRAM



Sifive E31 core complex



Features of E31 Core Complex

- E31 Core complex
- Developer :Sifive
- Includes :
    - E31 core : RV32IMAC
    - PLIC : Handle global Interrupts
    - CLINT
    - I- cache
    - DTIM
    - Debug Module : JTAG supported
    - Internal Bus : Tile-link

Evaluation RTL of E31 core

Advantages

- Fully functional,synthesizable,verilog RTL
- Port Interfaces : AHB-Lite
- Number of Global Interrupts : 127

Limitations

- Obfuscated verilog
- Reduced system/peripheral port address space.
- Reduced DTIM

Clocking Restrictions

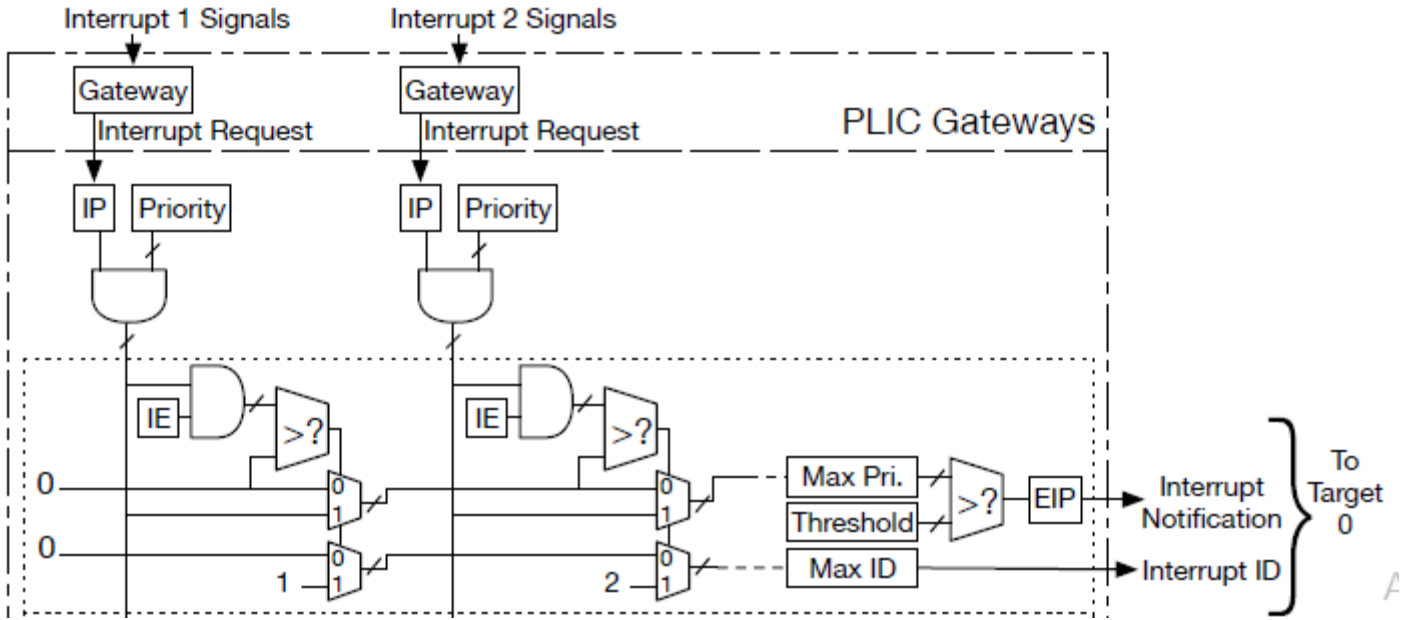
- Core_clock_0 rate must be equal to or higher than the periphery clock rate.
- Local_interrupt_0 must be synchronized to core_clock.
- Global_interrupts must be synchronized to clock.
- Real time clock : 32.768 KHz
- Reset must be synchronized to clock.

$$\text{Core_clock_0} \geq \text{clock} > 2 * \text{rtc_toggle}$$

Memory Map

Base Address	Top Address	Component	Description
0x0000_0000	0x0000_1000	Debug Controller	
0x0200_0000	0x0201_0000	CLINT	Core local Interrupt
0x0800_0000	0x0800_4000	ITIM	16 KB
0x0C00_0000	0x1000_0000	PLIC	Interrupt Controller Global interrupts :127
0x2000_0000	0x2000_07FF	GPIO	Peripheral port (8 KB)
0x2000_0800	0x2000_0FFF	Timer	
0x2000_1000	0x2000_1FFF	SPI	
0x4000_0000	0x4000_1FFF	SRAM (8KB)	System Port
0x8000_0000	0x8000_4000	DTIM	16 KB

PLIC


Interrupt 1 Signals
Interrupt 2 Signals
Gateway
Interrupt Request
Gateway
Interrupt Request
PLIC Gateways
IP
Priority
IP
Priority
IE
>?
IE
>?
0
0
1
2
Max Pri.
Threshold
Max ID
>?
EIP
Interrupt Notification
Interrupt ID
To Target 0

Features of PLIC

- The platform-level interrupt controller (PLIC) receives interrupt signals from the peripheral devices .
- Prioritizes the interrupts for service by the core.
- The E31 Core Complex has 127 interrupt sources.
- These signals are positive-level triggered.

PLIC Interrupt Sources

Interrupt number	Interrupt Sources
0	No interrupt
1 - 32	GPIO_0 - GPIO_31
33	Timer
34 - 127	Unused, tied to logic 0

Local Interrupt Sources

- Includes the standard software interrupts and timer interrupts.
- Local interrupts do not pass through the PLIC.
- Can be serviced quickly since there will be minimal latency between the source and the servicing hart.
- no arbitration is required to determine which hart will service the request, and the servicing hart.

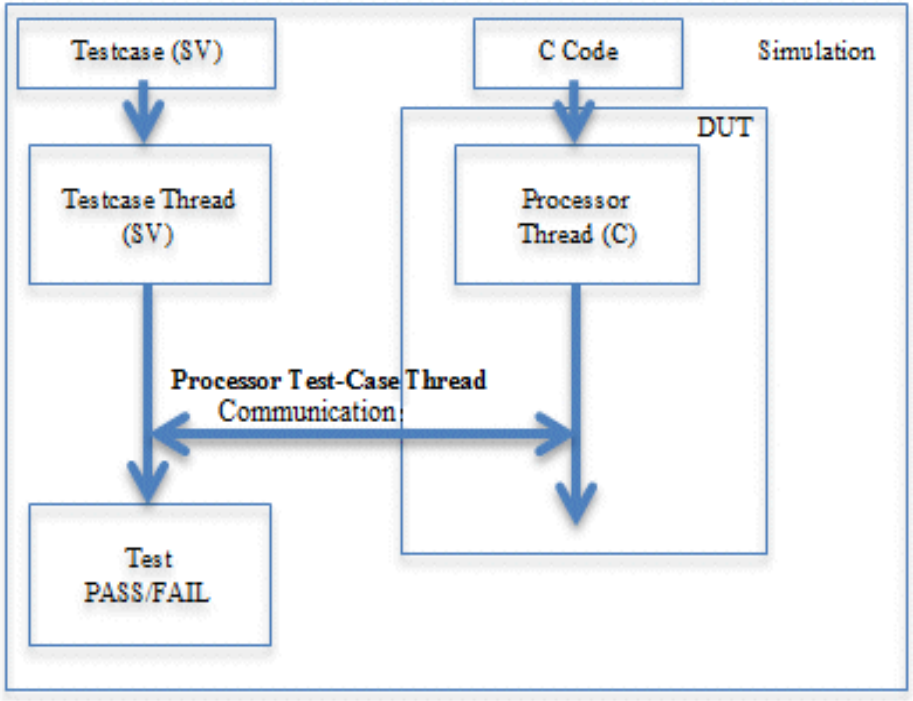
Interfaces of E31 Evaluation RTL

Top-level ports:

Function	Signal name	Width
Bus & periphery clock	clock	
Reset	reset	
Post-silicon debug mode	debug_psd_test_mode	
Post-silicon debug reset	debug_psd_test_mode_reset	
JTAG Debug	debug_systemjtag_*	
JTAG reset	debug_ndreset	
JTAG active	debug_dmactive	
Core clock	core_clock_0	
Reset vector address	reset_vector_0	32b
Realtime clock	rtc_toggle	
Global Interrupts	global_interrupts	128b
Local Interrupts	local_interrupts_0	16b
Connect AHB-Lite slaves	sys_port_ahb_0_*	32b
Connect AHB-Lite slaves	periph_port_ahb_0_*	32b
Connect AHB-Lite masters	front_port_ahb_0_*	32b

VERIFICATION

Block diagram for Processor Verification



Steps Performed

- Generation of c test cases
- The C test would get compiled and converted to a hex file.
- The hex file is preloaded to the memory.
- This preload gets called after reset deassertion.
- The System Verilog test bench could use sequences for reset and initialization.
- The processor in the SOC will fetch the hexcode from memory and execute same.

Freedom Studio SDK

- Freedom Studio is an integrated developer environment which can be used to write and debug software targeting SiFive based processors.

- Freedom Studio is bundled with a pre-built RISC-V GCC Toolchain, example programs, and documentation.

RISCV TOOLCHAIN

- riscv64-unknown-elf-gcc
  - To compile the C testcase
- riscv64-unknown-elf-objcopy
  - To convert the elf file to hexcode
- riscv64-unknown-elf-objdump
  - To dump the elf file
- riscv64-unknown-elf-as
  - To compile assembly file

TESTS PERFORMED

- System Port Verification
  - SRAM Memory Test
- Peripheral Port Verification
  - GPIO Test
  - SPI Test
  - Timer Test
- Front port Verification
  - I2C Verification
- Interrupt Verification

CHALLENGES FACED

- No transactions after connecting core to design module.
  - Reset should be more than 16 clock cycles
- After the fetch of 16 instruction, no transaction was happening.
  - The hex code was not in little endian format
- ‘X’ propagation in Hburst and I-cache and most of the other signals.
  - declare all MACRO needed to reset the RTL
  - RANDOMIZE_MEM_INIT
  - RANDOMIZE_REG_INIT
  - RANDOMIZE_GARBAGE_ASSIGN
  - RANDOMIZE_INVALID_ASSIGN

CHALLENGES FACED

- Instruction was not executing properly due the presence of random value in I-cache along with the hex code
  - All the memories were initialized to zero
- Read operation was not happening properly.
  - Delta delay issue while generating clock in the testbench
- During function calls in the c test, the hex code generated exceeded the SRAM memory space.
  - Added a flag “nostdlib” in gcc to avoid the c library calls and removed the “g” flag which contains debugging information.

THANK YOU

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